

Static Gesture Classification — Run Report

Generated 2026-05-13 11:31:59

Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic · 35 trials
Training data path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic · 35 trials
Training data path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic · 36 trials
Training data path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic · 35 trials
Training data path 5	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic · 35 trials
Training data path 6	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic · 24 trials
Training data path 7	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic · 35 trials
Training data path 8	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic · 35 trials
Training data path 9	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic · 36 trials
Training data path 10	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic · 35 trials
Total training paths	10 · 341 trials total
Report output dir	/home/jestin/ThesisRepo/ML/NewReports/DynamicReports

Sensor selection

Hands	left, right
Segments	palm_mid, thumb, index, middle, ring, pinky, wrist
Modalities	Accel
Sensor columns	78 channels
Classes	7: Double_Lower, Double_Nothing, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll

Preprocessing, features & split

Resample steps	90
Butterworth	on · cutoff 10.0 Hz · order 4 · fs 30.0 Hz
Normalisation	minmax
Feature mode	stats+fft → feature dim 4212
Split / seed / CV	random_state=42 · CV: LOUO (10 folds, every user held out once)
Sample counts	train (post-aug)=341 · LOUO-evaluated samples=341

Augmentation (training only)

Enabled	no
Copies per train trial	4
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.9, 1.1)
Time warp	off · range=(0.9, 1.1)

Classifiers — LOUO cross-validation

Algorithm	CV mean (LOUO, 10 folds)	CV std	CV min	CV max
Random Forest	0.8704	0.0998	0.6857	1.0000

Per-class CV F1 (across all LOUO folds)

Class	Random Forest	Support
Double_Lower	0.923	50
Double_Nothing	0.667	46
Double_Pistol_Recoil	0.845	50
Double_Raise	0.900	50
Double_Wiggle	0.929	49
Double_cmere	0.917	51
Drum_Roll	0.860	45
macro avg	0.863	341

Per-user accuracy (each user as held-out fold)

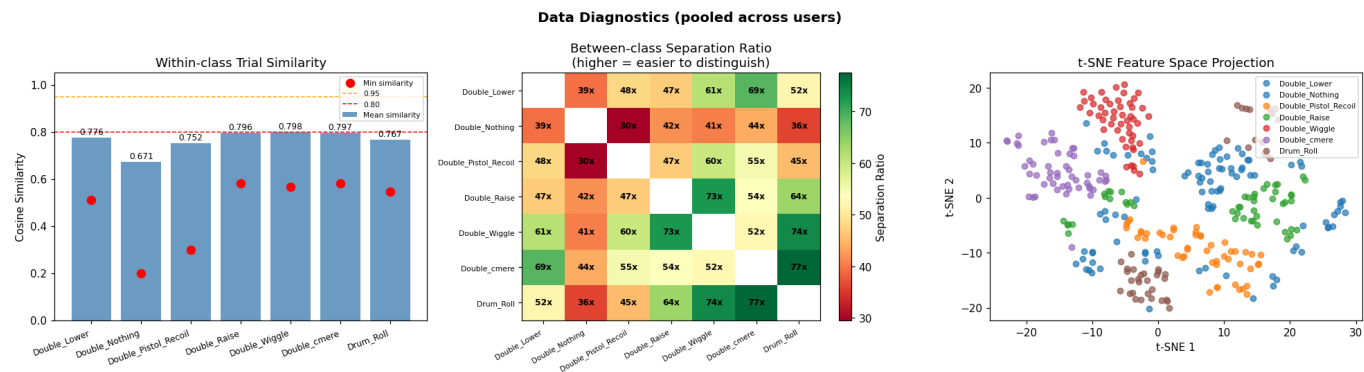
User	n trials	Random Forest
1_Alan	35	1.000
2_Alex	35	0.686
3_Anghad	36	0.972
4_Daniel	35	0.886
6_Harry	35	0.914
8_Jestin	24	0.958
11_Mansh	35	0.743
12_Marcus	35	0.771
14_StephenV2	36	0.917
17_Raquel	35	0.857

Mean (across users)		0.870
Std (across users)		0.100

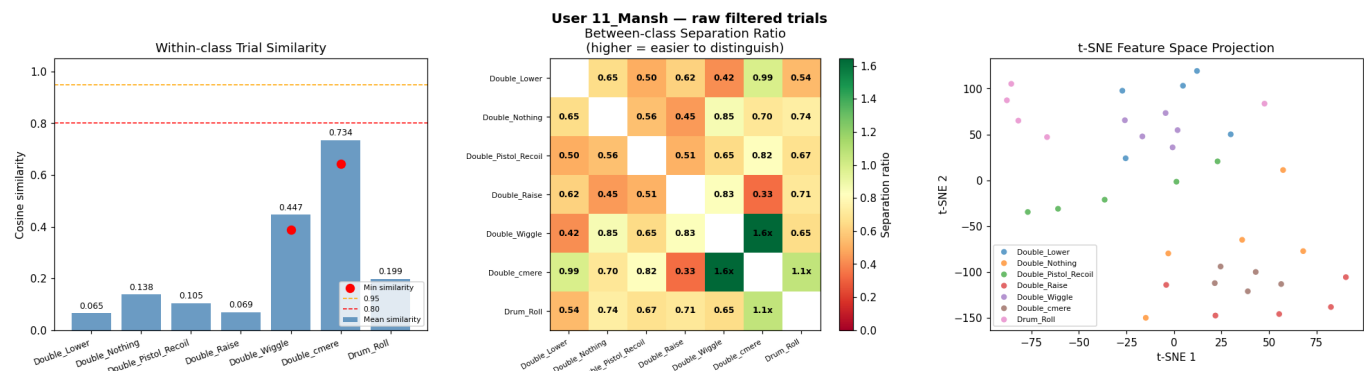
Appendix — Data Diagnostics

21 figures: pooled across users + one per detected user (raw filtered trials and extracted-feature views).

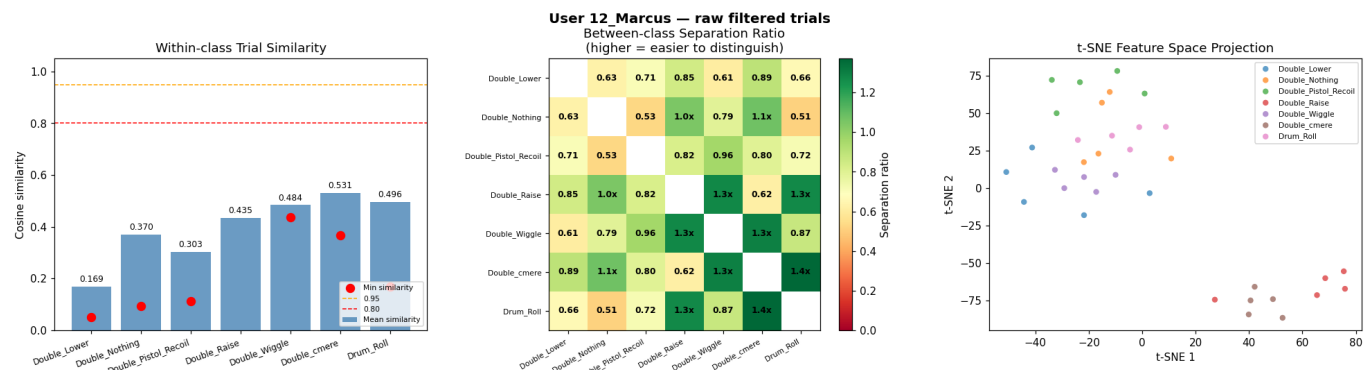
Pooled diagnostics — extracted features (across all users)



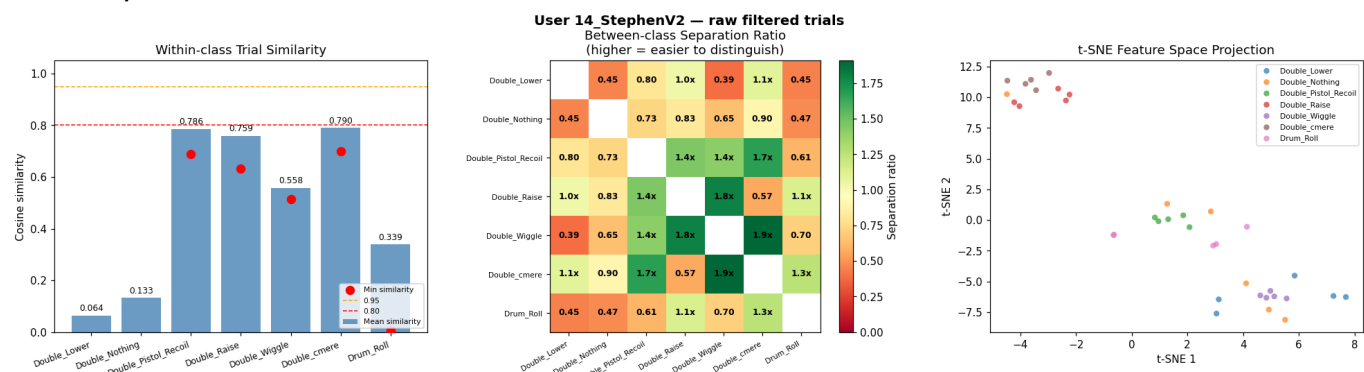
User 11_Mansh — raw filtered trials



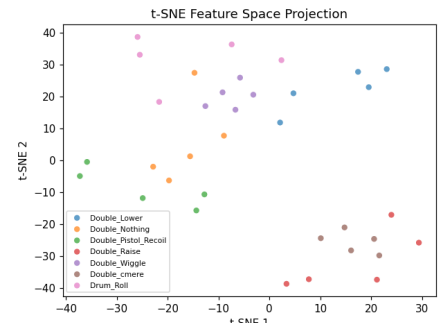
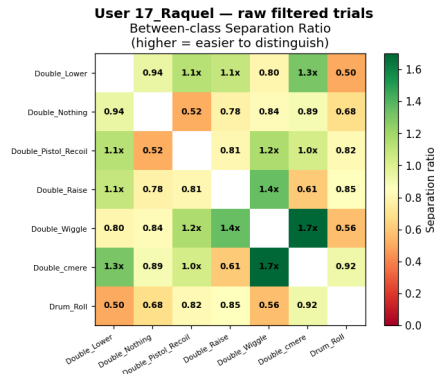
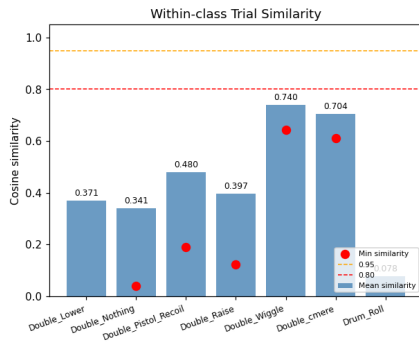
User 12_Marcus — raw filtered trials



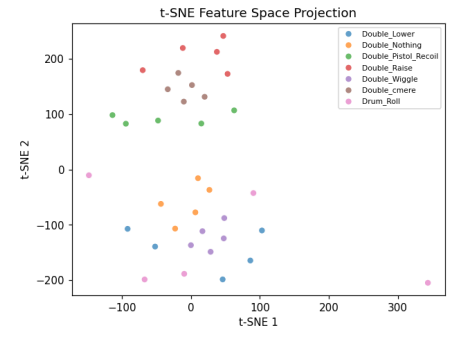
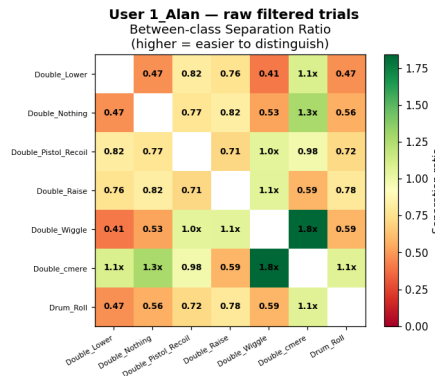
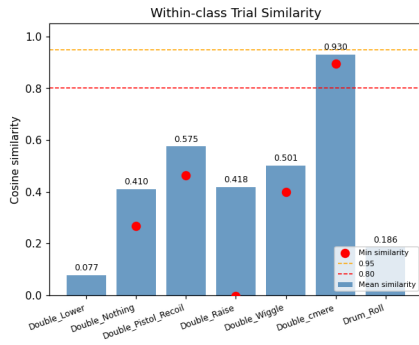
User 14_StephenV2 — raw filtered trials



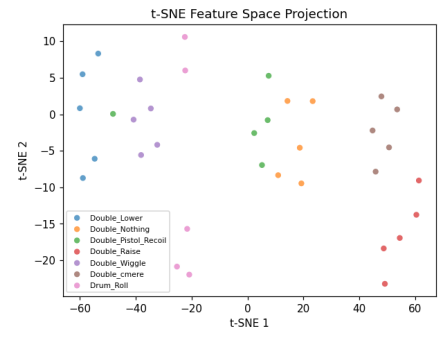
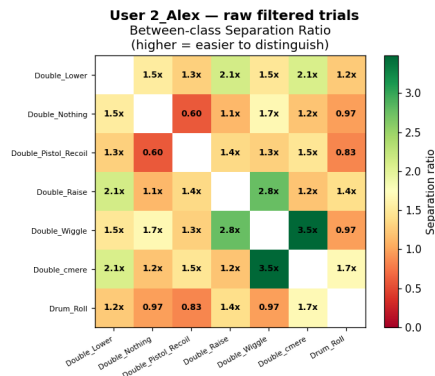
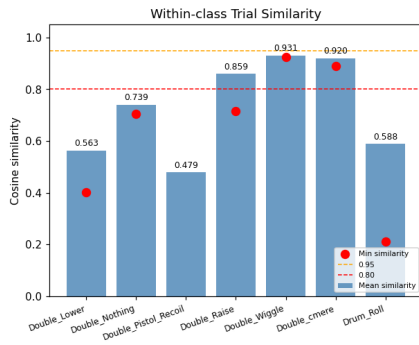
User 17_Raquel — raw filtered trials



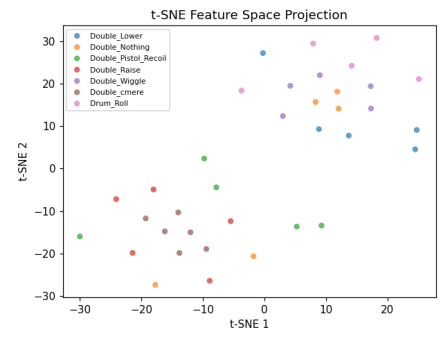
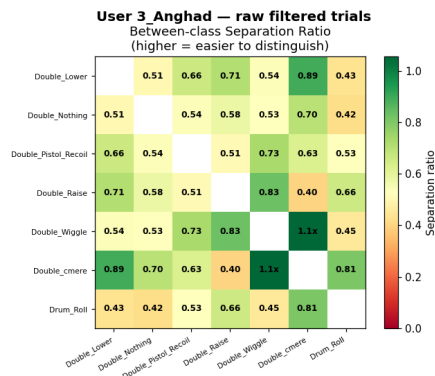
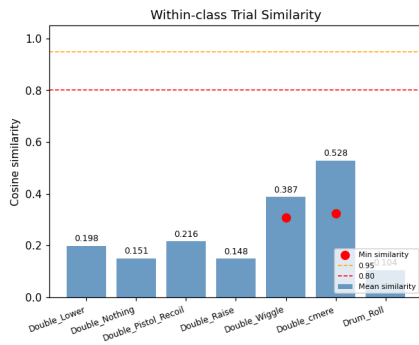
User 1_Alan — raw filtered trials



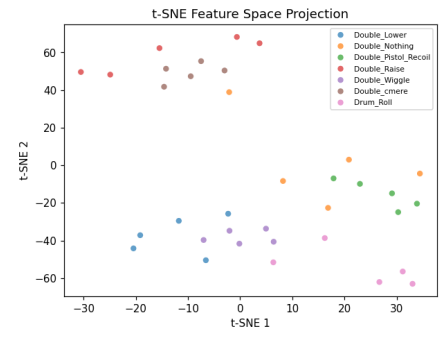
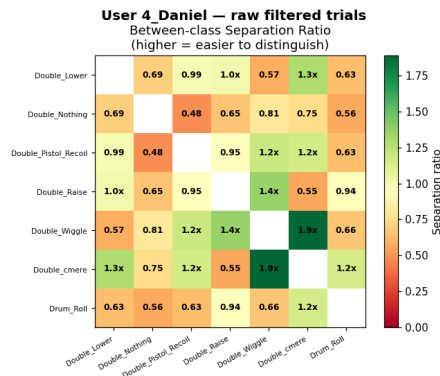
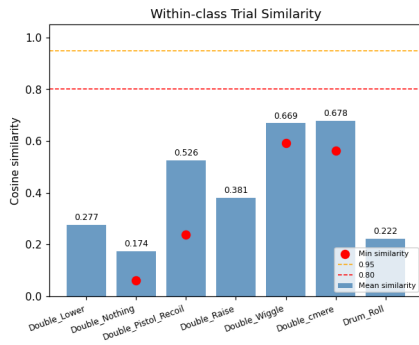
User 2_Alex — raw filtered trials



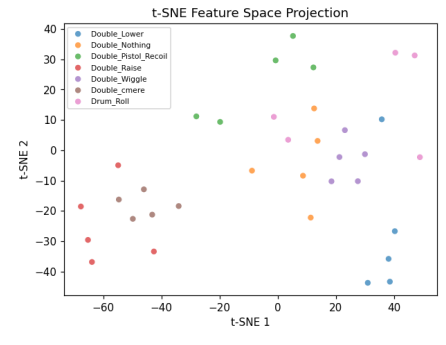
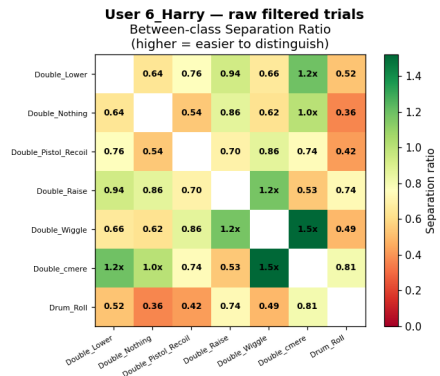
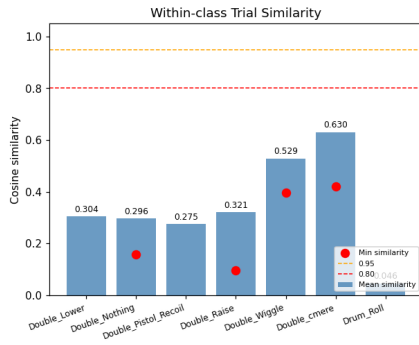
User 3_Anghad — raw filtered trials



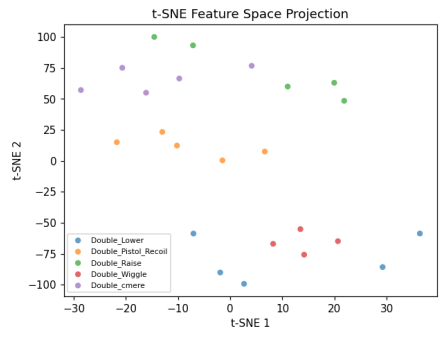
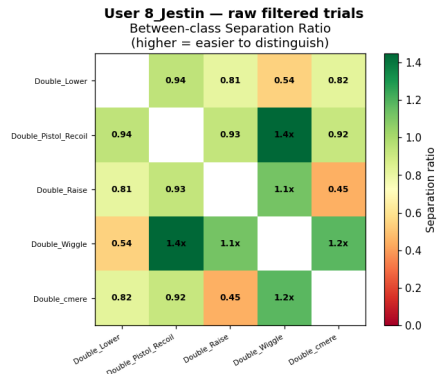
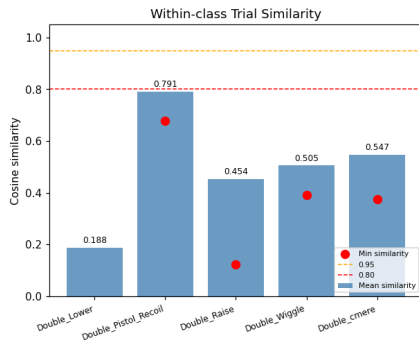
User 4_Daniel — raw filtered trials



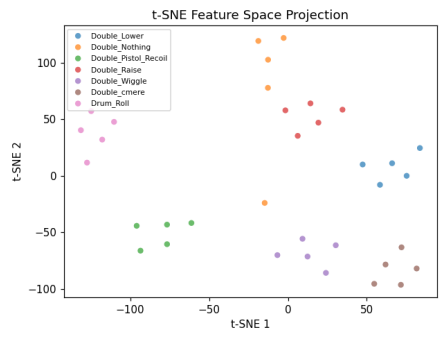
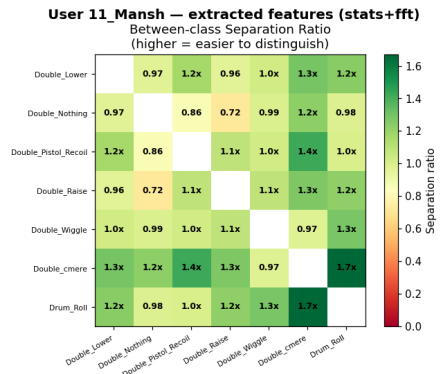
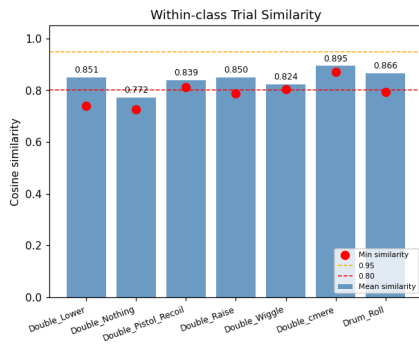
User 6_Harry — raw filtered trials



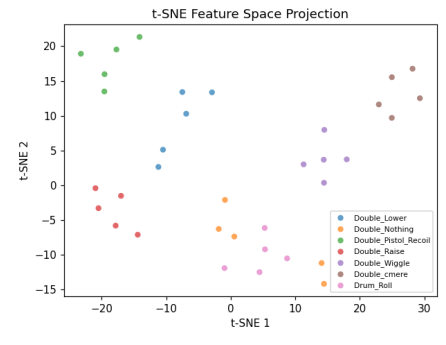
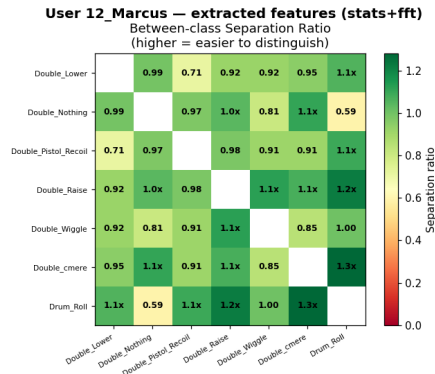
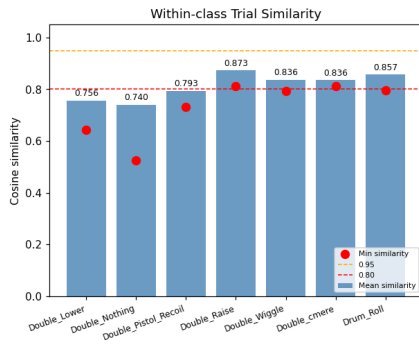
User 8_Jestin — raw filtered trials



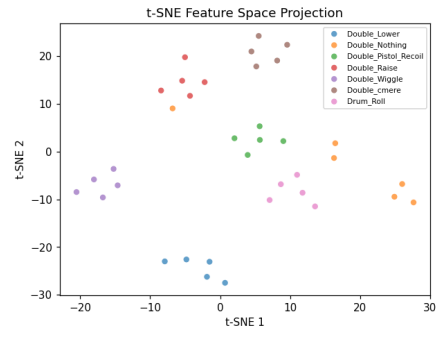
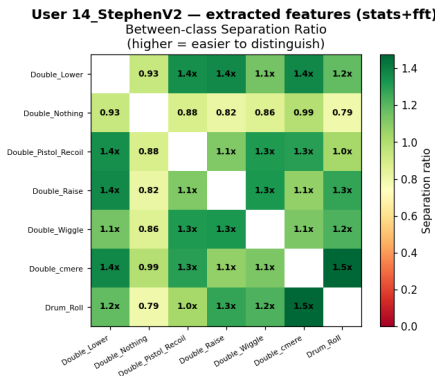
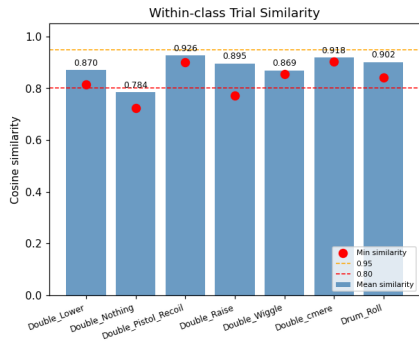
User 11_Mansh — extracted features (stats+fft)



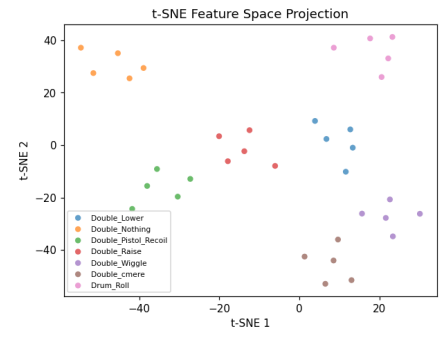
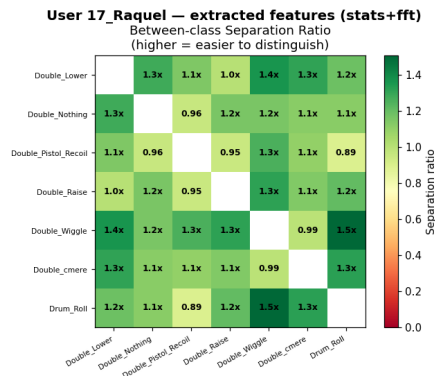
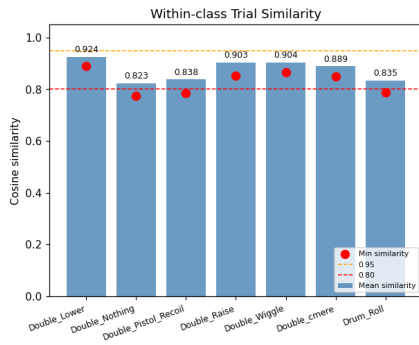
User 12_Marcus — extracted features (stats+fft)



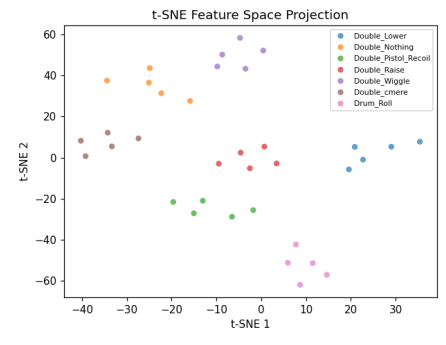
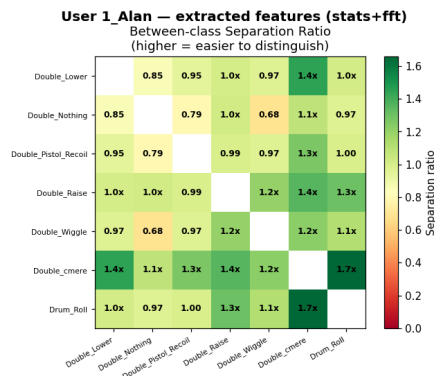
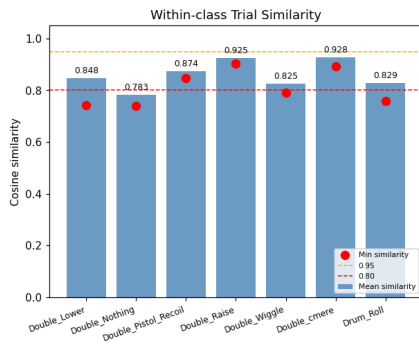
User 14_StephenV2 — extracted features (stats+fft)



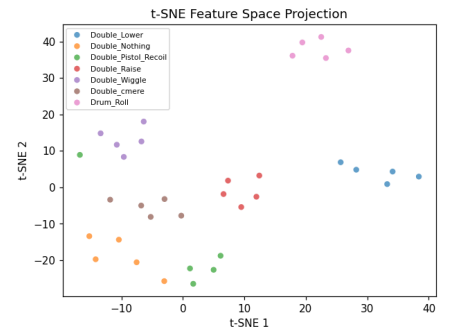
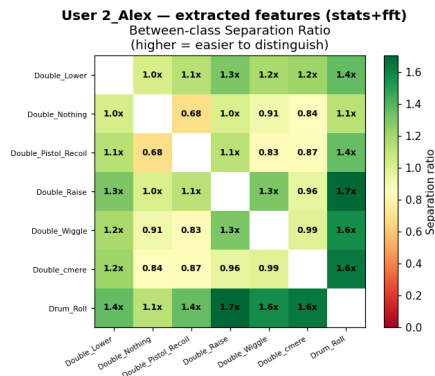
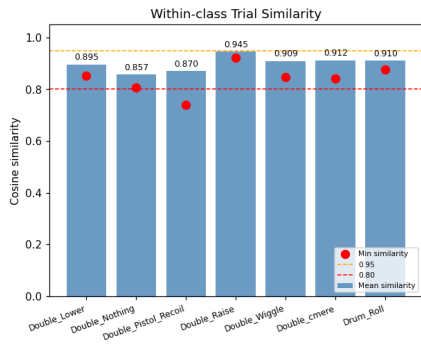
User 17_Raquel — extracted features (stats+fft)



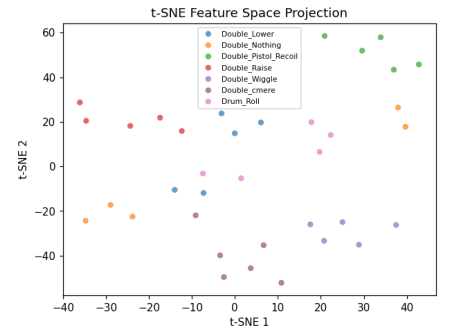
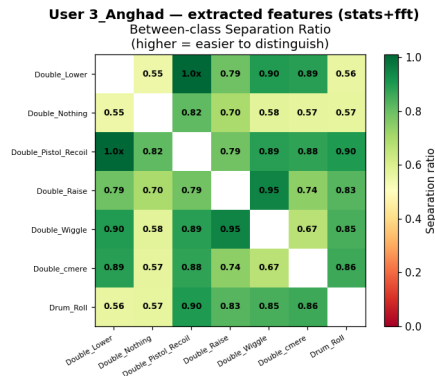
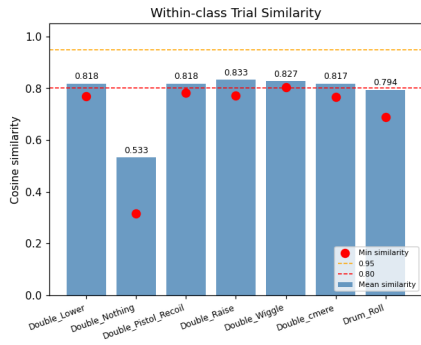
User 1_Alan — extracted features (stats+fft)



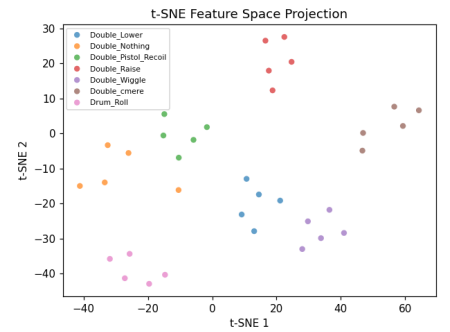
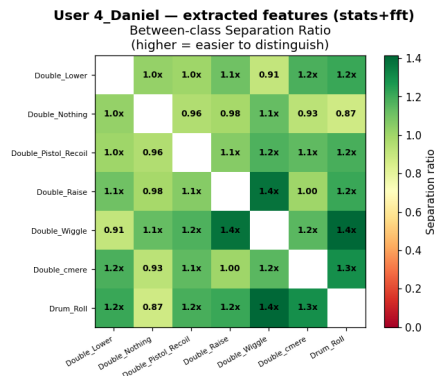
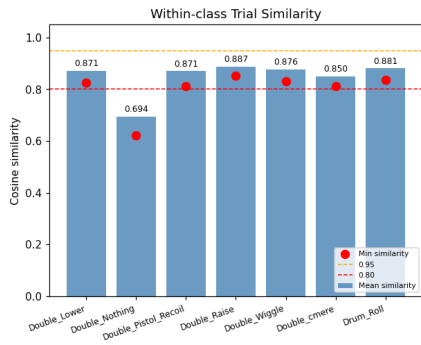
User 2_Alex — extracted features (stats+fft)



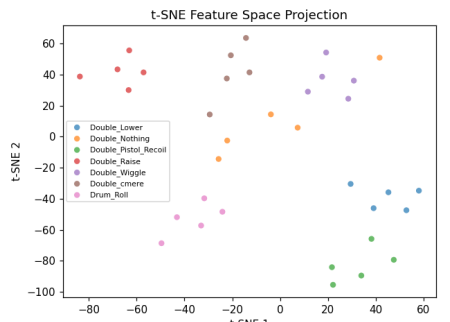
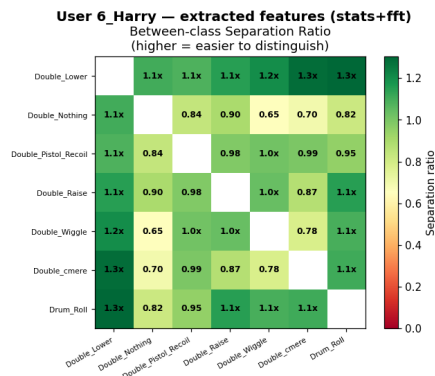
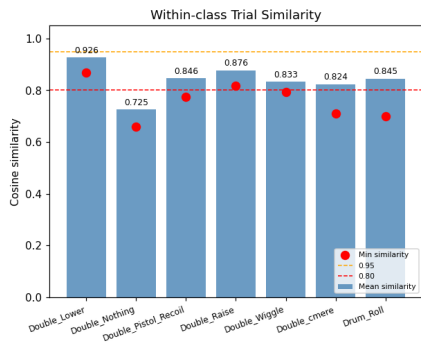
User 3_Anghad — extracted features (stats+fft)



User 4_Daniel — extracted features (stats+fft)



User 6_Harry — extracted features (stats+fft)



User 8_Jestin — extracted features (stats+fft)

